# Lab Assignment 4: Customer Churn Prediction

CSAI2017P: Applied Machine Learning (Lab)

Experiment 5 (Units V–VI) • CO2, CO4

**Instructor:** Dr. Tofik Ali **Due:** announced in the lab
**Student Name:** ____________________ **SAP ID:** ______________
**Batch:** __________ **Lab quiz:** LQ4, after submission

## Objective

- Build a classification pipeline over mixed numeric and categorical features.
- Read a confusion matrix and choose a metric that matches the business cost.
- Handle class imbalance and show what it changes.

## Datasets used in this assignment

**A2 (Telco Customer Churn).**
Full descriptions, sources and loading snippets for all anchor datasets are in `AnchorDatasets.pdf`, alongside this brief.

## Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

**Submit:**

- Notebook: `AML_LA04_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

| Section | Level | Choice rule | Marks |
|---|---|---|---|
| A | Easy | Attempt any 2 of 3 (2 marks each) | 4 |
| B | Medium | Attempt any 1 of 2 (4 marks each) | 4 |
| C | Hard | Compulsory | 2 |
| **Total** | | | **10** |

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

## Section A (Easy) — attempt any 2 of 3 $(2 \times 2 = 4)$

**A1. Encode and fit** **[2 marks]**

*Dataset: A2. Target:* `Churn`.

(a) Build a `ColumnTransformer` with one-hot encoding for categoricals and scaling for numerics.

(b) Fit a logistic regression inside the pipeline and report test accuracy.

(c) State the churn rate in the data and compare it with your accuracy in two lines.

**A2. Confusion matrix** **[2 marks]**

*Dataset: A2.*

(a) Print the confusion matrix and the classification report for your A1 model.

(b) State the number of churners the model missed.

(c) In two lines, say which error costs the company more: a missed churner, or a retention offer sent to someone who was staying.

**A3. Threshold moving** **[2 marks]**

*Dataset: A2.*

(a) Take `predict_proba` from your model and re-classify at thresholds 0.5, 0.35 and 0.25.

(b) Report precision and recall on the churn class for each.

(c) State which threshold you would ship and why, in two lines.

## Section B (Medium) — attempt any 1 of 2 $(1 \times 4 = 4)$

**B1. Model comparison** **[4 marks]**

*Dataset: A2.*

(a) Train logistic regression, a decision tree (`max_depth=5`) and a random forest on the same pipeline.

(b) Report accuracy, precision, recall, F1 and ROC-AUC for each in one table.

(c) Name the best model for this problem and defend the choice in 4–6 lines, referring to the cost argument.

**B2. Class imbalance** **[4 marks]**

*Dataset: A2.*

(a) Retrain your best model with `class_weight='balanced'`, and separately with random oversampling of the minority class.

(b) Report accuracy, recall and F1 for the baseline and both treatments in one table.

(c) Explain in 4–6 lines what each treatment did to the accuracy-recall trade-off, and whether it was worth it.

## Section C (Hard) — compulsory (2)

**C1. Which features carry the signal** **[2 marks]**

*Dataset: A2.*

(a) Report the ten most important features from your best model (coefficients or feature importances).

(b) In 4–6 lines, say whether any of them would be unavailable at the moment you actually need the prediction.

*End of Lab Assignment 4*

*Sources:* Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.